

WCE Deep Learning Project

Complete Project Explanation Guide

How to explain everything to the sir | Where is every code + image

1. What is This Project — The Big Picture

In simple words: We trained deep learning models to look at images from a tiny camera that goes inside someone's stomach (WCE = Wireless Capsule Endoscopy) and automatically detect diseases like polyps, ulcers, bleeding, and other abnormalities.

The main challenge: The dataset has 47,238 images but 72.7% of them are 'Normal' (healthy tissue). Only 10 images are Ampulla of vater (rarest disease). A stupid model that always says 'Normal' gets 72% accuracy — but it would MISS every single disease. That is clinically dangerous.

KEY INSIGHT

We cannot use accuracy as our metric. We use MACRO F1 — which gives equal weight to all 14 disease classes. A model that misses polyps always gets a low Macro F1, no matter how well it predicts Normal.

2. The Dataset (Section 2 of Report)

Dataset 1 — Kvasir-Capsule (Main)

47,238 WCE images, 14 classes, captured from real patients. The imbalance ratio is 3433.8:1 — Normal has 34,338 images, Ampulla has only 10.

Class	Count	Type	Clinical Meaning
Normal clean mucosa	34,338	Majority	Healthy stomach lining
Ileocecal valve	4,189	Majority	Junction of small & large intestine
Pylorus	1,529	Majority	Stomach exit valve
Angiectasia	866	Majority	Abnormal blood vessels — can bleed
Ulcer	854	Majority	Open sore in stomach lining
Erosion	506	Majority	Surface damage to stomach lining
Erythema	159	MINORITY	Redness — sign of inflammation
Polyp	55	MINORITY	Abnormal tissue growth — cancer risk
Blood - hematin	12	MINORITY	Old blood — serious finding

Class	Count	Type	Clinical Meaning
Ampulla of vater	10	MINORITY	Rarest — junction of bile & pancreatic duct

Dataset 2 — KVASIR v2 (Cross-Dataset Test)

8,000 images, 8 classes (different camera, conventional colonoscopy). We used 3,000 overlap images: 2,000 Polyp + 1,000 Ulcer. We applied our trained models WITHOUT retraining to see if they generalise.

HOW TO EXPLAIN TO SIR

Sir, we have two datasets. D1 is the main Kvasir-Capsule (47K WCE images, 14 classes, severe imbalance). D2 is KVASIR v2 (8K colonoscopy images). We evaluate our trained models on D2 without any retraining to test generalisation.

3. Three Imbalance Handling Settings (Section 3 of Report)

This is one of our core contributions. We compare 3 ways to handle the severe class imbalance:

Setting	Training Data	Strategy	Train Size
S1 — No Handling	Raw data as-is	Nothing done, use class weights in loss	33,066
S2 — Under-sample	Cap majority at 200	Remove excess Normal/ majority images	2,165
S3 — US + Augment	Cap majority + augment minority	Cap to 200, then augment all to 1000	14,000

S3 Augmentation Pipeline

For S3, we augment every class to exactly 1,000 images using:

- Random Horizontal + Vertical Flip (p=0.5 each)
- Random Rotation ±30 degrees
- Random Affine: translate 20%, scale 80-120%, shear 10°
- Colour Jitter: brightness ±30%, contrast ±30%, saturation ±20%
- Random Grayscale (p=0.05), Random Perspective (p=0.3)

CRITICAL LESSON: Shared Fixed Test Set

Before this fix: each setting had its own test set. S1 test had 5,134 Normal images (72.7%) — biased towards majority. S3 test had balanced images. This made comparison INVALID.

THE FIX

We hold out 15% of raw data (7,086 images) as a fixed test set BEFORE any resampling. All 3 settings (S1, S2, S3) are evaluated on the EXACT SAME 7,086 test images. This is the correct methodology.

4. Focal Loss — Our Loss Function (Section 3.2 of Report)

Standard Cross-Entropy Loss: every image contributes equally. Problem: model sees 34,338 Normal images — it learns to always say Normal (lowest loss).

Focal Loss solution: down-weight easy/correctly-classified samples. Formula:

$$L_{\text{focal}} = -\alpha_t \times (1 - p_t)^\gamma \times \log(p_t)$$

- p_t = predicted probability of correct class
- $\gamma = 2.0 \rightarrow$ hard, misclassified samples get higher weight
- α_t = per-class weight based on inverse frequency

MOST IMPORTANT DISCOVERY

For S1 (raw data): use Focal Loss WITH class weights (α). For S3 (balanced data): use Focal Loss WITHOUT class weights ($\alpha=\text{None}$). Why? S3 is already balanced physically. Adding weights on top makes the model over-predict rare classes — Normal recall drops to 0%. This was the bug causing S2/S3 to fail in early runs.

5. Models We Trained (Section 3.3 of Report)

We trained 8 models total: 6 CNN + 2 Transformer. All use ImageNet pre-trained weights (transfer learning).

CNN Models (6 models = 3 architectures $\times$ 2 settings S1 and S3)

Model	Parameters	What makes it special	Our modification
EfficientNetB0	4.3M	Compound scaling: balances depth, width, resolution	Last 30% unfrozen, custom head: Dropout→Linear(256)→ReLU→Linear(14)
MobileNetV3-Small	1.5M	Designed for mobile, uses squeeze-and-excitation blocks	Last 30% unfrozen, custom head — Best S1 model: F1=0.814
ResNet50	25.6M	Deep residual connections prevent vanishing gradients	Layer4 unfrozen, custom head — Good generalisation

Transformer Models (2 models, trained on S3 only)

Model	Parameters	How it works	Key setting
ViT-B/16 (Vision Transformer)	86.5M	Splits image into 196 patches (16×16). Each patch is a 'token'. 12 transformer layers with multi-head self-attention.	AdamW lr=1e-4, full fine-tune, 20 epochs
Swin-T (Swin Transformer)	28.3M	Hierarchical shifted-window attention. Processes local windows of patches, then shifts windows to capture cross-boundary patterns.	AdamW lr=1e-4, full fine-tune, 20 epochs — BEST MODEL

WHY TRANSFORMER

Sir asked us to explore transformer models. ViT and Swin are state-of-the-art vision models. Key difference from CNN: CNN uses sliding convolutional filters (local patterns). Transformer uses self-attention (each patch looks at all other patches globally). This global view helps capture disease patterns spread across the image.

6. Classification Results (Section 5 of Report)

All 8 Models on Shared Test Set (7,086 images)

Model	Setting	Acc(w)	F1(w)	Rec(macro)	F1(macro)	Rank
Swin-T	S3	0.9146	0.9211	0.9696	0.8331	 BEST

Model	Setting	Acc(w)	F1(w)	Rec(macro)	F1(macro)	Rank
ViT-B/16	S3	0.9008	0.9104	0.9524	0.8069	2nd
ResNet50	S3	0.8826	0.8922	0.9561	0.8095	3rd
MobileV3	S3	0.8808	0.8892	0.9441	0.7653	4th
EfficientNetB0	S3	0.8695	0.8855	0.9460	0.8075	5th
MobileV3	S1	0.8366	0.8517	0.9587	0.8143	Best CNN S1
EfficientNetB0	S1	0.7086	0.7336	0.9550	0.6597	—
ResNet50	S1	0.6581	0.6904	0.9465	0.6378	—

How to Explain These Numbers to Sir

Acc(w) = Weighted Accuracy

This is MISLEADING. A model predicting always 'Normal' gets 72.7% Acc(w). Useless clinically. We still report it for comparison with other papers.

F1(macro) = Our Primary Metric

Each of 14 classes contributes equally. Swin F1(macro)=0.8331 means: averaged across ALL 14 classes, it gets 83.3% F1. This is excellent for a 14-class severely imbalanced dataset.

Rec(macro) = Macro Recall

How many sick patients did we catch? Swin Rec(macro)=0.9696 means we correctly identify 96.96% of cases on average. All 4 minority classes have 0% miss rate — we NEVER miss Polyp, Erythema, Blood-hematin, or Ampulla.

S3 vs S1 — Did Balancing Help?

Yes. For EfficientNetB0: S1 F1(macro)=0.6597, S3 F1(macro)=0.8075 — that is +22.4% improvement. For ResNet50: S1 F1(macro)=0.6378, S3 F1(macro)=0.8095 — +26.9% improvement. Balancing works.

Swin vs CNN — Did Transformers Help?

Best CNN on S3: ResNet50 F1(macro)=0.8095. Swin F1(macro)=0.8331 — +2.9% better. More importantly, Swin Rec(macro)=0.9696 vs ResNet50 Rec(macro)=0.9561 — Swin catches more sick patients.

7. False Sample Analysis — Per-Class Miss Rate (Appendix of Report)

For Swin-T (best model), we analysed which classes it misses most:

Class	Total	Missed (FN)	Miss Rate	Type
Ileocecal valve	628	65	10.4%	Majority
Normal clean mucosa	5,151	499	9.7%	Majority
Pylorus	229	15	6.6%	Majority
Erosion	76	4	5.3%	Majority
Angiectasia	130	5	3.8%	Majority
Ampulla of vater	2	0	0.0% ✓	MINORITY
Erythema	24	0	0.0% ✓	MINORITY
Blood - hematin	2	0	0.0% ✓	MINORITY
Polyp	8	0	0.0% ✓	MINORITY

KEY POINT FOR SIR

All 4 minority classes (the rare/dangerous diseases) have 0% miss rate. The model NEVER misses Polyp, Erythema, Blood-hematin, or Ampulla. The hardest class is Ileocecal valve (10.4% miss) — these are visually similar to Normal tissue.

8. Anomaly Detection (Section 6 of Report)

Sir asked us to explore anomaly detection: can we detect 'something unusual is here' without knowing the exact disease? We implemented 5 methods and compared them.

What is the Anomaly Detection Task?

Binary decision: Normal (class 0) vs Anomaly = everything else (class 1). We evaluate using AUROC (area under ROC curve) — 1.0 = perfect, 0.5 = random guessing.

Method	Type	AUROC	AUPRC	How it works
Binary Detector	Supervised	0.9993	0.9988	Separate EfficientNetB0 trained to say Normal vs Not-Normal. Uses BCELoss.
One-Class SVM	Unsupervised	0.9590	0.9287	Train SVM on Normal features only. At test time: distance from Normal boundary = anomaly score.
K-Means Centroid	Unsupervised	0.9448	0.8857	Find centroid of all Normal feature vectors. Distance from centroid = anomaly score.
Isolation Forest	Unsupervised	0.9377	0.8820	Random decision trees. Anomalies are isolated faster (fewer splits needed).
Autoencoder	Unsupervised	0.6587	0.3631	Train ConvAE on Normal images only. High reconstruction error = anomaly.
Score-based softmax	Supervised	0.3452	0.2092	1 - max(softmax probability). Lower confidence = anomaly. FAILED — see below.

Explaining Each Method to Sir

Binary Detector
(AUROC=0.9993)

Best result. We trained a dedicated model: input=WCE image, output=Normal or Anomaly. This is supervised because it needs the Normal/Anomaly labels to train.

OCSVM
(AUROC=0.9590) —
BEST
UNSUPERVISED

We extract 1280-dimensional feature vectors from our S3 EfficientNetB0. Then we train OCSVM ONLY on Normal class features. At test time, if features are far from the learned Normal boundary, it is an anomaly. No disease labels needed at all.

K-Means / Isolation Forest
(AUROC ~0.94)

Similar idea: extract features from Normal images, learn what Normal looks like, flag anything far from Normal as anomaly. These are classical unsupervised ML methods applied to deep features.

Autoencoder (AUROC=0.6587) — Why it underperformed

We trained a convolutional autoencoder on 24,036 Normal images. It learns to compress and reconstruct Normal images. But GI tissue has similar low-level textures across all classes — so the AE can also reconstruct anomaly images reasonably well. Hence poor anomaly discrimination.

Score-based 1-max(softmax) (AUROC=0.3452) — Why it FAILED

This method assumes Normal class will have very high softmax probability (since it dominates training). But Swin on S3 is a BALANCED model — it gives equal probability to all 14 classes. So Normal doesn't dominate softmax. AUROC=0.3452 is worse than random (0.5) — meaning anomalies get LOW anomaly scores. This method only works with S1 (imbalanced) models.

Key Finding for Sir

OCSVM achieves AUROC=0.9590 with NO disease labels. Binary Detector achieves 0.9993 WITH labels. The gap is only 0.04. This proves: if you have good deep features from a classifier, you can do excellent anomaly detection without labelling any anomaly data.

9. GradCAM — What Are the Models Looking At? (Section 7 of Report)

GradCAM = Gradient-weighted Class Activation Maps. It shows WHICH part of the image the model used to make its decision.

CNN GradCAM Results

All 6 CNN models produce clear, localised heatmaps (red/yellow spots on the disease region):

- Polyp: Red/yellow activation covers the polyp body and its edges — correct clinical region
- Erythema: Tight activation on the reddened mucosal patch — exactly what a doctor looks at
- Blood-hematin: Covers the dark blood deposit area — model found it correctly
- Ampulla of vater: Activates on the distinctive tissue fold pattern

MobileNetV3 shows the TIGHTEST, most focused activations. S3 models show slightly tighter activations than S1 models — balanced training made the model more spatially precise.

Why Swin and ViT GradCAM Shows Uniform Purple

IMPORTANT — Tell This to Sir

GradCAM works by looking at gradients in convolutional feature maps. CNN has spatial feature maps (e.g. 7×7 grid of features). Transformer has token-based representations (196 flat tokens) — NO spatial feature maps. When GradCAM hooks into Swin, it gets uniform gradients because the transformer processes all patches globally. The uniform purple map is EXPECTED and NOT a failure. Swin still achieves the best F1=0.833.

What We Use Instead for Transformers

- ViT: Attention Rollout — multiplies attention matrices across all 12 layers to show which image regions the CLS token attends to
- Swin: Stage-3 spatial feature activation — hooks the last spatial stage, takes mean activation across channels, shows which patches are activated

10. t-SNE Feature Space Visualisation (Section 8 of Report)

t-SNE reduces high-dimensional features (1280-dim) to 2D for visualisation.

CNN vs Swin t-SNE

Plot: tsne_CNN_vs_Swin.png. Blue dots = EfficientNetB0 features. Red dots = Swin features. If they form separate clusters, the two architectures learned different representations. If they mix, they learned similar features but through different paths.

D1 vs D2 t-SNE

Plot: tsne_D1vsD2.png. Blue circles = Kvasir-Capsule (D1). Red crosses = KVASIR v2 (D2). If Polyp/Ulcer images from D2 land near Polyp/Ulcer images from D1 — the features TRANSFER across datasets. If D1 and D2 form separate clusters, the features do not generalise.

HOW TO EXPLAIN TO SIR

Sir, t-SNE is a dimensionality reduction technique. We take the 1280-dimensional feature vectors learned by the model and project them to 2D. If disease classes form tight, separate clusters it means the model learned discriminative features. If D2 images cluster near D1 images of the same class, the model generalises across different endoscopy equipment.

11. Cross-Dataset Generalisation — D2 Evaluation (Section 5.2 of Report)

We applied our D1-trained models to D2 (KVASIR v2) without any retraining. We evaluate only on the 3,000 images from overlapping classes (2,000 Polyp + 1,000 Ulcer).

The generalisation gap = D1 F1(macro) - D2 F1(macro). Smaller gap = better transfer.

STATUS

The exact D2 numbers are being generated by wce_phase3_final.py (currently running). The generalisation_gap.csv file in outputs_phase3 will have the final numbers. Fill into Table 5.2 of the report when available.

12. Ablation Study (Section 9 of Report)

Focal Loss Weight Impact

Configuration	Normal Recall	Macro F1	What happened
S3 + weights max=100	0.00 (collapsed)	~0.43	Model predicted only rare classes for everything
S3 + weights max=10	0.01 (near zero)	~0.43	Same collapse, just slightly less severe
S3 + NO weights	0.84 (correct)	0.81	Balanced dataset handles imbalance; no extra weights needed
S1 + weights max=10	0.61 (correct)	0.66	Imbalanced data needs weights to learn rare classes

Augmentation Target Impact

AUG_TARGET	Macro F1	Normal Recall	Training set size
200 (= cap, no augmentation)	0.43	0.01	2,165 (no augmentation happened)
500	0.75	0.80	~8,000
1000 (final)	0.81	0.84	14,000

13. CODE LOCATION MAP — Where is Every Script

This section is for when sir asks 'show me the code for X'. Use this to find the exact file and line.

What sir asks about	File name on HPC	Full HPC path	Key section in file
Data loading, S1/S2/S3 split	5_wce_major_final.py	/home/rahuldixit/aksh/dl_project/5_wce_major_final.py	Lines ~60-130: BalancedDS class, IndexDS class, split logic
Focal Loss (Phase 1)	5_wce_major_final.py	/home/rahuldixit/aksh/dl_project/5_wce_major_final.py	Lines ~145-165: FocalLoss class, focal_s1 vs focal_s3
EfficientNetB0 model	5_wce_major_final.py	/home/rahuldixit/aksh/dl_project/5_wce_major_final.py	Lines ~170-185: build_effnet() function
MobileNetV3 model	5_wce_major_final.py	/home/rahuldixit/aksh/dl_project/5_wce_major_final.py	Lines ~187-197: build_mobilenet() function
ResNet50 model	5_wce_major_final.py	/home/rahuldixit/aksh/dl_project/5_wce_major_final.py	Lines ~199-210: build_resnet50() function
ViT-B/16 model	5_wce_major_final.py	/home/rahuldixit/aksh/dl_project/5_wce_major_final.py	Lines ~212-217: build_vit() function
Swin Transformer model	5_wce_major_final.py	/home/rahuldixit/aksh/dl_project/5_wce_major_final.py	Lines ~219-224: build_swin() function
Training loop (all models)	5_wce_major_final.py	/home/rahuldixit/aksh/dl_project/5_wce_major_final.py	Lines ~228-295: train() function, CosineAnnealingLR + ReduceLROnPlateau
GradCAM implementation	6_wce_major_final.py	/home/rahuldixit/aksh/dl_project/6_wce_major_final.py	Lines ~115-175: GradCAM class, generate(), overlay()

What sir asks about	File name on HPC	Full HPC path	Key section in file
ViT Attention Rollout	6_wce_major_final.py	/home/rahuldixit/aksh/dl_project/6_wce_major_final.py	Lines ~180-240: get_vit_attention() function, MultiheadAttention hook
Autoencoder (fixed, no sigmoid)	6_wce_major_final.py	/home/rahuldixit/aksh/dl_project/6_wce_major_final.py	Lines ~255-310: ConvAE class, NormalDS class, AE training loop
K-Means centroid anomaly	6_wce_major_final.py	/home/rahuldixit/aksh/dl_project/6_wce_major_final.py	Lines ~360-375: KMeans fit on norm_feats, distance calculation
Isolation Forest anomaly	6_wce_major_final.py	/home/rahuldixit/aksh/dl_project/6_wce_major_final.py	Lines ~377-385: IsolationForest(200, contamination=0.2)
One-Class SVM anomaly	6_wce_major_final.py	/home/rahuldixit/aksh/dl_project/6_wce_major_final.py	Lines ~387-397: OneClassSVM(rbf, nu=0.1)
D2 cross-dataset loading	7_wce_phase3_final.py	/home/rahuldixit/aksh/dl_project/7_wce_phase3_final.py	Lines ~100-150: D2DS class, D2_TO_D1 mapping
D2 evaluation + gap table	7_wce_phase3_final.py	/home/rahuldixit/aksh/dl_project/7_wce_phase3_final.py	Lines ~155-210: d2_results loop, gap_rows calculation
t-SNE D1 vs D2	7_wce_phase3_final.py	/home/rahuldixit/aksh/dl_project/7_wce_phase3_final.py	Lines ~215-275: extract_features, TSNE(2, perplexity=30)
t-SNE CNN vs Swin	7_wce_phase3_final.py	/home/rahuldixit/aksh/dl_project/7_wce_phase3_final.py	Lines ~280-340: swin_hook_fn, combined features, TSNE plot

What sir asks about	File name on HPC	Full HPC path	Key section in file
Swin feature activation vis	7_wce_phase3_final.py	/home/rahuldixit/aksh/dl_project/7_wce_phase3_final.py	Lines ~345-395: swin_feature_vis(), hook on features[-3]
Final 6-panel summary figure	7_wce_phase3_final.py	/home/rahuldixit/aksh/dl_project/7_wce_phase3_final.py	Lines ~400-510: 2x3 subplots, all results combined

14. IMAGE LOCATION MAP — Where is Every Figure

Use this when sir asks to see a specific figure. These are the HPC paths.

Figure in Report	File name	HPC directory	Description
Fig: Class distribution	P1_distribution.png	outputs_final/	Bar chart + pie chart of 14-class distribution
Fig: All model F1 comparison	FINAL_SUMMARY.png	outputs_phase3/	6-panel summary: all results
Fig: Generalisation gap	D1_vs_D2_generalisation.png	outputs_phase3/	D1 vs D2 F1 bar chart + gap bar chart
Fig: Confusion matrix (Swin)	S3_Swin_CM.png	outputs_phase2/	14×14 confusion matrix for best model
Fig: Per-class ROC (Swin)	S3_Swin_ROC_classes.png	outputs_phase2/	14 ROC curves, one per class
Fig: GradCAM EffNetB0 S1	gradcam_S1_EffNetB0.png	outputs_phase2/	GradCAM for 4 minority classes, S1
Fig: GradCAM EffNetB0 S3	gradcam_S3_EffNetB0.png	outputs_phase2/	GradCAM for 4 minority classes, S3
Fig: GradCAM MobileV3 S1	gradcam_S1_MobileV3.png	outputs_phase2/	Sharpest CNN GradCAM (S1)
Fig: GradCAM MobileV3 S3	gradcam_S3_MobileV3.png	outputs_phase2/	Sharpest CNN GradCAM (S3)
Fig: GradCAM ResNet50 S1	gradcam_S1_ResNet50.png	outputs_phase2/	ResNet50 GradCAM (S1)
Fig: GradCAM ResNet50 S3	gradcam_S3_ResNet50.png	outputs_phase2/	ResNet50 GradCAM (S3)

Figure in Report	File name	HPC directory	Description
Fig: GradCAM Swin (uniform)	gradcam_S3_Swin.png	outputs_phase2/	Purple uniform — GradCAM not valid for Swin
Fig: ViT Attention Rollout	vit_attention_maps.png	outputs_phase2/	ViT attention on minority classes: 3-column view
Fig: Swin Feature Activation	swin_feature_activation.png	outputs_phase3/	Correct Swin interpretability (viridis map)
Fig: Unsupervised anomaly ROC	anomaly_unsupervised_ROC.png	outputs_phase2/	4 unsupervised methods ROC comparison
Fig: Autoencoder reconstructions	autoencoder_reconstructions.png	outputs_phase2/	Normal vs anomaly reconstruction + error maps
Fig: t-SNE CNN vs Swin	tsne_CNN_vs_Swin.png	outputs_phase3/	Feature distribution: CNN blue, Swin red
Fig: t-SNE D1 vs D2	tsne_D1vsD2.png	outputs_phase3/	Feature distribution: D1 circles, D2 crosses
Fig: Training curves Swin	S3_Swin_curves.png	outputs_final/	Loss, Val Macro F1, LR schedule over 20 epochs
Fig: Training curves ViT	S3_ViT_curves.png	outputs_final/	Loss, Val Macro F1, LR schedule over 20 epochs
Fig: False positives Swin	false_positives_S3_Swin.png	outputs_phase2/	10 worst misclassified images for Swin

15. SAVED MODEL PATHS — Where Are the .pth Files

Model	.pth file name	Directory
S1 EfficientNetB0	S1_EffNetB0_best.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/
S1 MobileNetV3	S1_MobileV3_best.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/
S1 ResNet50	S1_ResNet50_best.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/
S3 EfficientNetB0	S3_EffNetB0_best.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/
S3 MobileNetV3	S3_MobileV3_best.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/
S3 ResNet50	S3_ResNet50_best.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/
S3 ViT-B/16	S3_ViT_best.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/
S3 Swin-T (BEST)	S3_Swin_best.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/

Model	.pth file name	Directory
Binary Anomaly Detector	binary_detector.pth	/home/rahuldixit/aksh/dl_project/saved_models_final/
Autoencoder (fixed)	autoencoder_fixed.pth	/home/rahuldixit/aksh/dl_project/outputs_phase2/

16. How to Verbally Explain the Project to Sir

Opening — 2 minutes

"Sir, our project is on automatic disease detection in Wireless Capsule Endoscopy images. A patient swallows a tiny camera capsule that captures 50,000+ images as it goes through the stomach. Currently doctors review all these manually — very time-consuming. We built deep learning models to automate this."

The Challenge — 1 minute

"The main challenge is class imbalance. Our dataset has 47,238 images but 72.7% are Normal. Only 10 images are Ampulla — the rarest disease. A naive model always says Normal and gets 72% accuracy but misses every disease. We use Macro F1 as our metric — it treats all 14 classes equally."

What We Did — 3 minutes

"We tried three approaches to fix the imbalance: S1 uses raw data with weighted Focal Loss. S2 under-samples majority classes to 200 images each. S3 under-samples AND augments minority classes to 1,000 images each — perfectly balanced."

"We trained 8 models: EfficientNetB0, MobileNetV3, ResNet50 (these are CNN), and ViT and Swin Transformer (as sir suggested). All use ImageNet pre-trained weights. Transformer models use AdamW with lower learning rate because they are more sensitive."

Key Results — 2 minutes

"Swin Transformer on balanced dataset: F1(macro)=0.8331, Recall(macro)=0.9696. All 4 rare disease classes: 0% miss rate. The model never misses Polyp, Blood-hematin, Erythema, or Ampulla. Swin beats all CNN models."

"For anomaly detection: our One-Class SVM achieves AUROC=0.9590 without ANY anomaly labels — it only trains on Normal images and flags anything unusual. The supervised binary detector gets 0.9993. Gap of only 4%."

Interpretability — 1 minute

"GradCAM on CNN models shows the model correctly activates on the lesion regions: polyp body, blood deposits, inflamed patches. For Swin Transformer, GradCAM doesn't apply (no convolutional maps), so we use Attention Rollout for ViT and stage-level feature activation for Swin."

Cross-Dataset — 1 minute

"We also tested on KVASIR v2 — a different dataset from conventional colonoscopy — without retraining. We evaluated on 3,000 Polyp and Ulcer images. This shows whether our model generalises beyond WCE data. [Report D2 F1 numbers when available]"

17. Quick Answers If Sir Asks Specific Questions

If sir asks...	Answer to give
Why Focal Loss?	Standard cross-entropy: every Normal image contributes equally. Model learns 'always say Normal'. Focal Loss: down-weight correctly classified easy samples. Misclassified rare classes keep high gradient. Gamma=2 means $(1-p)^2 \approx 0$ for easy Normal, stays high for missed polyps.
Why no class weights for S3?	S3 is physically balanced (1000 per class). If we also add 10x weight for rare classes, the model is double-penalised and ignores Normal completely. Normal recall drops to 0%. Weights are only for S1 (still imbalanced raw data).
Why Swin better than ViT?	Swin uses hierarchical feature maps and shifted-window attention — processes local patterns first then builds global understanding. ViT treats all 196 patches equally from the start, which is harder to fine-tune on small datasets. Swin is also faster.
Why OCSVM works so well?	The S3 EfficientNetB0 backbone learned to separate Normal from disease in 1280-dimensional space. OCSVM simply draws a boundary around the Normal cloud in this space. Any point outside = anomaly. The boundary generalises well because the features are discriminative.
Why autoencoder underperformed?	GI tissue has similar texture patterns (mucosal surface) across all classes. The autoencoder learns to reconstruct these textures well for both Normal and anomaly images. Without explicit anomaly labels, it cannot distinguish them.

If sir asks...	Answer to give
What is GradCAM?	Gradient-weighted Class Activation Maps. We compute the gradient of the predicted class score with respect to the last convolutional feature map. Positive gradients = features that pushed the prediction toward this class. We weight each feature map channel by its gradient and take the ReLU of the weighted sum.
Why purple map for Swin?	GradCAM needs convolutional spatial feature maps (7×7 or 14×14 grid). Swin's last stage produces patch embeddings, not spatial convolution maps in the traditional sense. The gradient hook captures token representations where spatial gradients are uniform. We use Swin's stage-3 feature activation instead — physically meaningful.
What is your test set?	Fixed 15% of raw data (7,086 images) held out BEFORE any resampling. Same test set for all 3 settings. Stratified split with random seed=42 for reproducibility.
What is imbalance ratio?	3433.8:1. Normal clean mucosa has 34,338 images, Ampulla of vater has 10 images. This means for every 1 Ampulla image there are 3,433 Normal images.
What GPU did you use?	NVIDIA H100 NVL, 100 GB VRAM, CUDA 12.8. Each full training run (8 models) took approximately 2 hours.

End of Explanation Guide — WCE Major Project